# Calculations report: TEST INPUT - readiness trace pump panel

Job number: TEST-001

**NOT APPROVED: issued for review only.**

The design rules are this software's defaults; the company has not confirmed them.

## Basis of calculation

- Design current Ib = P / (k U cos phi); breaker In >= Ib, cable Iz >= In (IEC 60364-4-43 §433.1; ABB Electrical Installation Handbook Vol. 2, §2.1-2.3).
- Current-carrying capacity from IEC 60364-5-52 Annex B, corrected for ambient temperature and grouping.
- Voltage drop from the origin, from Schneider EIG Fig. G28 (copper) and ABB §2.2.2 (aluminium), against IEC 60364-5-52 Annex G or the company's limits.
- Fault level at a sub-board through its feeder, IEC 60909-0 (cmax 1.10); breaking capacity at or above it.
- Earth fault disconnection, IEC 60364-4-41 §411.4.4: Zs x Ia <= 0.95 U0, Ia the top of the breaker's instantaneous band (IEC 60898-1).
- Short circuit along the cable, ABB §2.4: the far-end Ikmin trips the breaker at once; $k^2S^2$ from Table 1 for comparison with the breaker's let-through $I^2t$.

## Board PP1

400 V, 3-phase, TN-S · Ik 10 kA · Ze - ohm · Incomer C100 4P 10 kA

| Circuit | Ib A | Protection | Cable | L m | dU % | Zs ohm | Zs max | $k^2S^2$ $(kA)^2s$ |
|---|---|---|---|---|---|---|---|---|
| Panel lighting (test) | 1 | C10 1P 10 kA | 3G1.5 Cu PVC | 10 | 0.12 | - | 2.194 | 0.03 |
| Duty pump (test) | 15.2 | MA20 3P | 4G1.5 Cu PVC | 30 | 2.85 | - | - | 0.03 |
| Standby pump (test) | 56 | MA80 3P | 7G6 Cu PVC | 30 | 1.82 | - | - | 0.476 |
| Booster (test) | 10.4 | 25 A | 4G1.5 Cu PVC | 20 | 1.3 | - | - | 0.03 |

### Notes

- Group 1 breaker raised to 16 A, 1.6 x the 10 A breaker after it, for discrimination.
- Duty pump (test): Ir 15.2 A, the typical Ir of a 7.5 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate.
- Duty pump (test): dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2.
- Standby pump (test): Ir 56 A, the typical Ir of a 30 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate.
- Standby pump (test): star_delta starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 5: 400 V 50 kA Y/D Normal Type 2.
- Standby pump (test): star and delta contactors need a mechanical interlock; the motor is fed by six conductors (U1 V1 W1, U2 V2 W2).
- Booster (test): Ir 10.38 A from 5.5 kW shaft power at cos phi 0.85, efficiency 0.9; check it against the nameplate.

- Booster (test): ACS880-01-12A6-3 for normal duty at 30 °C, behind 25 A aR fuses (ABB ACS880-01 hardware manual (3AUA0000078093)). Use a screened, symmetrical motor cable, earthed 360° at both ends.

- Booster (test): the fuses need at least 75 A prospective short-circuit current at the board to operate fast enough.

- Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too.

- Leave 1 spare outgoing ways (20 %).

- Loads without a power factor were taken at cos phi 0.9.

- Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.

- No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze.

- Each cable's withstand $k^2S^2$ is in the cable list; compare it with the selected breaker's let-through $I^2t$ at 10 kA from its maker's curve (ABB handbook §2.4).

- Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.

- Circuit rules are this software's defaults, not confirmed by the company's engineers.

- Enclosure: none named, since 11 devices have no width (rail_widths_mm in the company settings).

| Prepared | Checked | Approved |
|---|---|---|
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